UniGe | MaLGa

MLCC 2026

Variable Selection and Sparsity

Cesare Molinari

MaLGa, UniGe-Dima

Outline

Variable Selection

Greedy Methods: (Orthogonal) Matching Pursuit

Convex Relaxations: LASSO & Elastic Net

Prediction and Interpretability

- In practice: beyond prediction, **interpretability** (detecting which factors allow prediction)
- Moreover, less variables may imply more precise prediction...

$\longrightarrow$ **Variable Selection!** (Occam's razor)

Example: price of the houses

Notation Recap

- **Linear Data Model (reality):** Training Set $\{(x_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}$ with

$$\langle w^*, x_i \rangle = y_i$$

  What is y_i? A **linear measurement** of w^*...

- **Linear Prediction Model (artificial):**

$$f_w(x) = \langle w, x \rangle$$

- Empirical Risk Minimization:

$$\underbrace{\begin{bmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{bmatrix}}_{X \in \mathbb{R}^{n \times d}} w = \underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}}_{y \in \mathbb{R}^n}; \quad \text{i.e. } Xw = y$$

What do we want?

Linear Prediction Model:

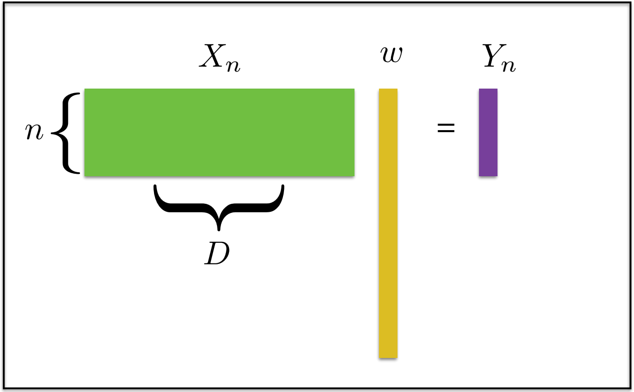
$$f_w(x) = \langle w, x \rangle = \sum_{j=1}^{d} w^j x^j$$

- Given $\{(x_i, y_i)\}_{i=1}^n$, which variables w^j are important for prediction?

Key assumption: the prediction rule is **sparse** (many w^j are zero)

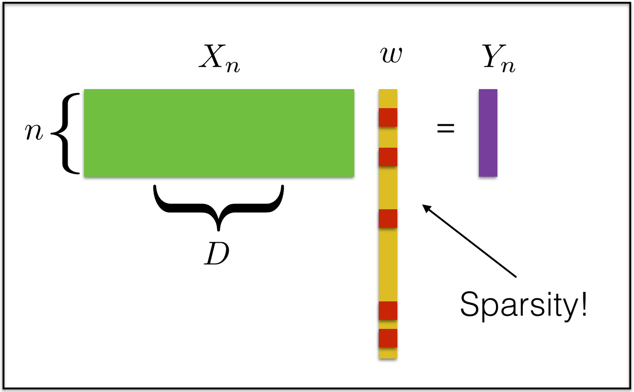
High-Dimensional Statistics with Sparsity

Estimating a linear model $\longrightarrow$ solving a linear system: $Xw = y$ with the info that w is **sparse**



High-Dimensional Statistics with Sparsity

Estimating a linear model $\longrightarrow$ solving a linear system: $Xw = y$ with the info that w is **sparse**



Estimating a linear model $\longrightarrow$ solving a linear system:

$$Xw = y$$

- If $n \gg d$, **overdetermined system** ($\nexists$ sol)
- If $n \ll d$, **underdetermined system** (∞ sols)

<u>Reminder:</u> generalized solution and pseudo-inverse

Compressed Sensing

Similar but different question: How many linear measurements to reconstruct $w \in \mathbb{R}^d$?

- In general...
- Compressed Sensing: if you know that w is **sparse**, you need way less! (not trivial...)

$\longrightarrow$ Compression!

Brute Force Approach

ℓ^0 Norm:

$$\|w\|_0 = |\{j \mid w^j \neq 0\}|$$

What do we want? To solve $Xw = y$ (approx) but also with $\|w\|_0$ small; so...

Square Loss + ℓ^0-Regularization:

$$\min_{w \in \mathbb{R}^d} \quad \frac{1}{2}\|Xw - y\|^2 + \lambda \|w\|_0$$

The Brute Force Approach is Hard

Square Loss + Regularization:

$$\min_{w \in \mathbb{R}^d} \quad \frac{1}{2}\|Xw - y\|^2 + \lambda \|w\|_0$$

- As hard as considering all possible subsets of variables (single, couples, triplets...)
- So, the computational complexity is **combinatorial (NP hard)**

Alternative approaches:

- Greedy Methods
- Convex Relaxation

Outline

Variable Selection

Greedy Methods: (Orthogonal) Matching Pursuit

Convex Relaxations: LASSO & Elastic Net

Greedy Methods

Initialize at $w_0 = 0$ (with residual $r_0 = y$ and index set $I_0 = \emptyset$); given I, w, r...

1. Find the variable j most correlated with the residual r
2. Update the index set I to include such variable
3. Update coefficient vector w
4. Update residual r as $r = y - Xw$

<u>Names:</u>

- **Forward Stage-Wise Regression**, in statistics
- **Matching Pursuit (MP)**, in signal processing

How we start

Notation:

- $w \in \mathbb{R}^d$: coefficient vector
- $r = y - Xw \in \mathbb{R}^n$: residual
- $I \subseteq [d]$: index set

Initizialization:

$$w_0 = 0, \qquad r_0 = y, \qquad I_0 = \emptyset$$

Normalization (of the columns):

$$\|X^j\| = 1$$

Next Variable

Select the variable j^* s.t. X^{j^*} best explains the residual r in **least-squares**:

- for each j, compute

$$v^j := \underset{v \in \mathbb{R}}{\arg\min} \quad \|r - X^j v\|^2, \qquad \text{so} \quad v^j = r^\top X^j$$

- then select

$$j^* = \underset{j=1,\dots,d}{\arg\min} \quad \|r - X^j v^j\|^2$$

Since

$$\|r - X^j v^j\|^2 = \|r\|^2 - (r^\top X^j),^2$$

we have

$$j^* = \underset{j=1,\dots,d}{\arg\max} \quad (r^\top X^j)^2$$

So: select the variable j s.t. X^j has largest **projection** on the residual!

Algorithm: Matching Pursuit

Given I, w, r

- Variable Selection:
$$j_+ \; = \; \underset{j=1,\dots,d}{\arg\max} \quad (r^\top X^j)^2$$
- Active Index Set Update:
$$I_+ \; = \; I \cup \{j_+\}$$
- Coefficients vector:
$$w_+ = w + v_+ e_{j_+}, \qquad \text{where} \quad v_+ = r^\top X^{j_+}$$
- Residual Update:
$$r_+ = r - X(v_+ e_{j_+})$$

Remark: at every iteration, one more variable

Orthogonal Matching Pursuit

At every iteration, **least-squares restricted to I variables**:

$$w_+ = \arg\min_{w \in \mathbb{R}^d} \|X M_{I_+} w - y\|^2,$$

where $M_I \in \mathbb{R}^{d \times d}$ is such that $(M_I w)^j = w^j$ if $j \in I$ and $(M_I w)^j = 0$ otherwise

(instead of computing only the coeff of the new variable, all the coeffs of the selected variables are recomputed)

Moreover, the residual update is replaced by

$$r_+ = y - X w_+$$

Restricted Isometry Constant (RIC): δ_s is the smallest quantity s.t.

$$(1-\delta_s)\|w\|_2^2 \leq \|Xw\|_2^2 \leq (1+\delta_s)\|w\|_2^2$$

for all s-sparse vectors w

(Compressed Sensing: there are ways to build X with $n << d$ and δ_s small, also random w.h.p.)

Theorem

If

$$\delta_{s+1} < \frac{1}{\sqrt{s}+1},$$

OMP recovers any s-sparse signal w^* from noise-free measurements $y = Xw^*$ in s iterations

Procedure

What do we want?

Recover a signal $w^* \in \mathbb{R}^d$ via few linear measurements with the a-priori info that is s-sparse...

- Build X with $n << d$ and $\delta_{s+1} < 1/(\sqrt{s}+1)$
- Perform the n linear measurements $y = Xw^*$
- Apply s iterations of OMP

Other Issues

For ML,

- You can not choose X
- Sparsity may not be known a-priori
- Noise in the output data y

Outline

Variable Selection

Greedy Methods: (Orthogonal) Matching Pursuit

Convex Relaxations: LASSO & Elastic Net

Convex Relaxation

ℓ^0-norm is replaced by the ℓ^1**-norm,** namely

$$\|w\|_1 = \sum_{j=1}^{d} |w^j|$$

Then... **LASSO**, in statistics; or **Basis Pursuit**, in signal processing:

$$\min_{w \in \mathbb{R}^d} \quad \frac{1}{2} \|Xw - y\|^2 + \lambda \|w\|_1$$

- convex but **not differentiable**!

How to solve it?

ISTA (iterative soft-thresholding algorithm):

$$w_{k+1} = S_{\lambda\gamma}\left(w_k - \gamma X^\top (X w_k - y)\right)$$

At each iteration a non-linear soft-thresholding operator is applied to a gradient step...

What is it? and...

Where does it come from?!

Convergence results?

ISTA

Iterative optimization algorithm:

$$w_{k+1} = S_{\gamma\lambda}(w_k - \gamma X^\top (X w_k - y))$$

- To ensure convergence, set
$$\gamma < \frac{2}{\|X\|^2}$$

- until a convergence criterion, e.g. $\|w_{k+1} - w_k\| \leq \epsilon$; or max number of iterations K_{max}

Splitting Methods

ISTA: the contribution of data-fitting and regularization are **split**...

- the argument of the soft thresholding operator corresponds to a step of gradient descent

$$w_k - \gamma X^\top (X w_k - y)$$

- the soft-thresholding operator depends only on the regularization and acts component wise

$$S_\alpha(u) = ||u| - \alpha|_+ \frac{u}{|u|}$$

$\longrightarrow$ **Sparsity:** the coefficients of the solution can be exactly zero

Lasso meets Tikhonov: Elastic Net

- Tikhonov solution is stable, but in general not sparse
- LASSO solution is sparse, but in general not stable

Elastic Net: for $\alpha \in [0,1]$,

$$\min_{w\in\mathbb{R}^d} \frac{1}{n}\sum_{i=1}^{n}(f_w(x_i)-y_i)^2+\lambda(\alpha\left\|w\right\|_1+(1-\alpha)\left\|w\right\|_2^2)$$

$\longrightarrow$ hybrid algorithm which interpolates between Tikhonov and LASSO

ISTA for Elastic Net

The ISTA procedure can be adapted to solve the elastic net problem, where the gradient descent step incorporates also the derivative of the ℓ^2 penalty term:

$$w_{k+1} = S_{\gamma\lambda\alpha}((1 - \gamma\lambda(1-\alpha))w_k - \gamma X^\top (Xw_k - y))$$

To ensure convergence, set

$$\gamma < \frac{2}{\|X\|^2 + \lambda(1-\alpha)}$$

Wrapping Up

Sparsity for variable selection and interpretable models

- Greedy Methods
- Convex Relaxations

Next Class

Unsupervised Learning: dimensionality reduction (and clustering)!